Manual Chapter : Configuring the BIG-IP System as a Reverse Proxy Server

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Configuring a Simple Intranet

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Configuring Content Adaptation for HTTP Requests

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Configuring HTTP Load Balancing with Source Address Affinity Persistence

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Configuring the BIG-IP System as a Reverse Proxy Server

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Configuring the BIG-IP System as a Reverse Proxy Server

Overview: URI translation and HTML content modification

For environments that use web servers, you might want your websites to appear differently on the external network than on the internal network. For example, you might want the BIG-IP system to send traffic destined for *http://www.siterequest.com/* to the internal server *http://appserver1.siterequest.com/* instead. Normally, this translation could cause some issues, such as the web server expecting to see a certain host name (such as for name-based virtual hosting) or the web server using the internal host name and/or path when sending a redirect to client systems. Fortunately, you can configure the BIG-IP system to solve these problems.

You can also configure the BIG-IP system to modify HTML content as needed after the system has performed the URI translation.

This implementation describes an example of URI translation and HTML content modification and then provides the tasks to implement this example.

About URI translation

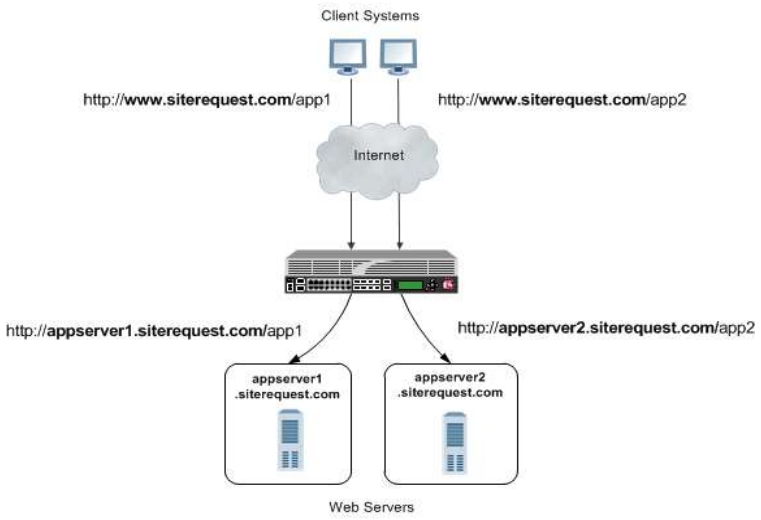
You can configure the BIG-IP system to perform URI translation on HTTP requests. Suppose that a company named `Siterequest` has a website *www.siterequest.com*, which has a public IP address and a registered DNS entry, and therefore can be accessed from anywhere on the Internet.

Furthermore, suppose that `Siterequest` has two application servers with private IP addresses and unregistered DNS entries, inside the company's firewall. The application servers are visible within the internal network as *appserver1.siterequest.com* and *appserver2.siterequest.com*.

Because these servers have no public DNS entries, any client system that tries to access one of these servers from outside the company network receives a no such host error.

As the illustration shows, you can prevent this problem by configuring the BIG-IP system to act as a reverse proxy server:

The BIG-IP system as a reverse proxy server for URI translation



In the example, the company `Siterequest` has decided to enable Web access to the internal application servers, without exposing them to the Internet directly. Instead, the company has integrated the servers with the web server *siterequest.com* so that *http://www.siterequest.com/sales* is mapped internally to *http://appserver1.siterequest.com/sales*, and *http://siterequest.com/marketing* is mapped internally to *http://appserver2.example.com/marketing*. This is a typical reverse-proxy configuration.

To configure the BIG-IP system to perform this translation, you create a Rewrite profile and configure one or more URI rules. A ***URI rule*** specifies the particular URI translation that you want the BIG-IP system to perform. Specifically, a URI rule translates the scheme, host, port, or path of any client URI, server URI, or both. A URI rule also translates any domain and path information in the Set-Cookie header of the response when that header information matches the information in the URI rule.

The Rewrite profile supports HTML and CSS content types only. To specify MIME types for HTML content, you can either create an HTML profile or accept the default values that the Rewrite profile uses, `text/html` and `text/xhtml`. For CSS content, only the `text/css` MIME type is supported.

Rules for matching requests to URI rules

The BIG-IP system follows these rules when attempting to match a request to a URI rule:

- A request does not need to match any entry. That is, if no entries match and there is no catch-all entry, then the Rewrite profile has no effect.
- Each request matches one entry only, which is the entry with the most specific host and path.
- If multiple entries match, then the BIG-IP system uses the entry with the deepest path name on the left side of the specified mapping.
- The BIG-IP system matches those requests that contain host names in URIs before matching requests that do not contain host names in URIs.
- The BIG-IP system processes the specified entries in the mapping from most-specific to least-specific, regardless of the order specified in the actual Rewrite profile.

About URI Rules

When creating a URI rule, you must specify the client and server URIs in these ways:

- When the URI is a path prefix only, the path must be preceded by and followed by a /, for example, */sales/*.
- When the URI contains more than the path prefix (such as, a host), the URI must also contain a scheme and must be followed by a /, for example, *http://www.siterequest/sales/*.

Introduction to HTML content modification

When you configure an HTML profile on the BIG-IP system, the system can modify HTML content that passes through the system, according to your specifications. For example, if you want the BIG-IP system to detect all content of type text/html and then remove all instances of the HTML img tag with the src attribute, you can configure an HTML profile accordingly, and assign it to the virtual server. The HTML profile ensures that the BIG-IP system removes those instances of the tag from any HTML content that passes through the virtual server.

Or, you can configure an HTML profile to match on a certain tag and attribute in HTML content when a particular iRule event is triggered, and then create an iRule that includes a command to replace the value of the matched attribute with a different attribute. The BIG-IP system includes several iRule commands that you can use when the Raise Event on Comment or Raise Event on Tag events are triggered. For more information on iRule commands related to HTML content modification, see the F5 Networks web site *http://devcentral.f5.com*.

HTML tag removal and replacement are just two of several HTML rules that you can configure to manipulate HTML content. An ***HTML rule*** defines the specific actions that you want the BIG-IP system to perform on a specified type HTML content.

Task summary for configuring system as reverse proxy server

The first step to configuring the BIG-IP system to act as a reverse proxy server is to create a Rewrite type of profile on the BIG-IP system and associate it with a virtual server. Note that each virtual server must have an HTTP profile. The Rewrite profile is designed for HTTP sites, as well as HTTPS sites where SSL is terminated on the BIG-IP system (that is, the virtual server references a Client SSL profile).

Creating a Rewrite profile to specify URI rules

To configure the BIG-IP system to perform URI translation, you create a ***Rewrite profile***, specifying one or more URI rules that associate a client-side path with a server-side URI. You also specify whether you want the URI translation to pertain to HTTP requests, responses, or both.

The Rewrite profile supports HTML and CSS content types only. To specify MIME types for HTML content, you can either create an HTML profile or accept the default values that the Rewrite profile uses, `text/html` and `text/xhtml`. For CSS content, only the `text/css` MIME type is supported.

1. On the Main tab, click **Local Traffic › Profiles › Services › Rewrite** .
   The Rewrite profile list appears.
2. Click **Create New Profile**.
   The Create New Profile Rewrite pop-up screen opens.
3. In the **Profile Name** field, type a name, such as `my_rewrite_profile`.
4. From the **Parent Profile** list, select `rewrite`.
5. From the **Rewrite Mode** list, select **URI Translation**.
6. On the left pane, click **URI Rules**.
   An empty text box appears for displaying client-server URI mappings that you specify.
7. Click **Add**.
8. From the **Rule Type** list, select **Both**.
9. In the **Client URI** box, type a client path, such as */sales/*.
10. In the **Server URI** box, type a server URI, such as *http://appserver1.siterequest.com/sales/*.

    You must include a scheme in the server URI that you specify.

    An example of a scheme is *http*.
11. Click **OK**.
    This displays a mapping of the specified client path to the associated server scheme, host, and path.
12. Click **Add** again.
13. From the **Rule Type** list, select **Both**.
14. In the **Client URI** field, type a client path, such as */marketing/*.

15. In the **Server URI** field, type a server URI, such as *http://appserver2.siterequest.com/marketing/*.

    You must include a scheme in the server URI that you specify.

    An example of a scheme is *http*.

16. Click **OK.**
    This displays a mapping of the specified client path to the associated server scheme, host, and path.

17. Click **OK.**

The BIG-IP system now includes two URI rules for performing URI translation on both requests and responses. For example, the host name in a request destined for *http://www.siterequest.com/sales/* will be translated to *http://appserver1.siterequest.com/sales/*, and the host name in a request destined for *https://www.siterequest.com/marketing/* will be translated to *http://appserver2.siterequest.com/marketing/*. A reverse translation occurs on any response.

Creating an HTML profile for tag removal

You create an HTML profile when you want the BIG-IP system to act on certain types of HTML content.

1. On the Main tab, click **Local Traffic › Profiles › Content › HTML .**
2. Click the **Create New Profile** button.
3. In the **Profile Name** field, type a name, such as `my_html_profile`.
4. From the **Parent Profile** list, select **/Common/html**.
5. On the left pane, click **HTML Rules.**
6. On the **Create New** button, click the right arrow.
7. Select **Remove Tag.**
   The Create New Remove Tag Rule box appears.
8. In the **Rule Name** field, type a name, such as `my_remove_img_tag_rule`.
9. Optionally, in the **Description** field, type a description of the rule, such as `Removes the img tag with the src attribute`.
10. On the left pane, click **Match Settings.**
11. In the **Match Tag Name** field, type the name of the tag that you want to remove from the HTML content.
    An example of a tag to specify is the HTML img tag.
12. In the **Match Attribute Name** field, type the name of the attribute associated with the tag that you specified for removal.
    An example of an attribute to specify is the src attribute for the img tag.
13. Click **OK.**
14. In the **Available Rules** list, locate the HTML rule that you want to enable, and select the adjacent check box.
15. Using the Move button, move the selected HTML rule to the **Selected Rules** list.
16. Click **OK.**

After creating this HTML profile, you can implement the HTML content modification by assigning the profile to the virtual server that is processing the associated HTTP traffic.

Creating pools for processing HTTP traffic

You can create two load balancing pools, and then create a policy that forwards certain HTTP traffic to one pool, and other HTTP traffic to another pool.

1. On the Main tab, click **Local Traffic › Pools .**
   The Pool List screen opens.

2. Click **Create.**
   The New Pool screen opens.
3. In the **Name** field, type a unique name for the pool.
4. Using the **New Members** setting, add each resource that you want to include in the pool:
   a. (Optional) In the **Node Name** field, type a name for the node portion of the pool member.
   b. In the **Address** field, type an IP address.
   c. In the **Service Port** field, type a port number, or select a service name from the list.
   d. (Optional) In the **Priority** field, type a priority number.
   e. Click **Add.**
5. Click **Finished.**
6. Repeat this task to create a second pool.

The new pools appear in the Pools list.

Creating a draft local traffic policy

You can use BIG-IP local traffic policy matching to direct traffic in accordance with rules, which are applied as determined by the specified strategy, conditions, and actions.

Local traffic policies that have been upgraded from BIG-IP software version 12.0, or earlier, appear in the Published Policies list.

1. On the Main tab, click **Local Traffic › Policies › Policy List .**
   The Policy List Page screen opens.
2. Click **Create.**
   The New Policy screen opens.
3. In the **Policy Name** field, type a unique name for the policy.
4. In the **Description** field, type a description for the policy.
5. From the **Strategy** list, select a matching strategy.
6. Click **Create Policy.**
   The policy is created and the Rules area appears.
7. In the Rules area, click **Create.**
8. In the Match all of the following conditions area, click **+.**
9. From the **Client SSL** list, select a condition type, and configure the applicable settings and available options.
10. Click **Add.**
11. In the Match all of the following conditions area, click **+** to add an additional condition, as necessary, and configure the applicable settings and available options.
12. In the Do the following when the traffic is matched area, click **+.**
13. From the **Enable** list, select an action type, and configure the applicable settings and available options.
14. Click **Add.**
15. In the Do the following when the traffic is matched area, click **+** to add an additional action, as necessary, and configure the applicable settings and available options.
16. Click **Save.**

The policy appears in the Draft Policies list.

Implementation results

After you perform the tasks in this implementation, the BIG-IP system can:

- Translate URIs according to the URI rules specified in the Rewrite profile.
- Modify specified HTML content according to the HTML rule specified in the HTML profile.
- Forward HTTP traffic to two different non-default pools according to a local traffic policy.

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